Lab - Bicep - Deploying an Azure Web App
To Begin with the Lab

Summary

This lab demonstrates how to use **Azure Bicep** to deploy an Azure Web App. You will create a Bicep template that provisions a Resource Group, App Service Plan, and Azure Web App. After deployment, you will verify the resources in the Azure Portal and clean up the environment.

Understanding

Azure Bicep is an Infrastructure as Code (IaC) language that simplifies Azure resource deployment. Instead of manually creating resources through the Azure Portal, you define the infrastructure in a .bicep file and deploy it using Azure CLI or PowerShell. Bicep is easier to read and write than ARM templates while providing the same capabilities.

An **Azure Web App** is a Platform as a Service (PaaS) offering that hosts web applications without requiring you to manage servers. By combining Bicep with Azure Web Apps, you can automate deployments, ensure consistency, and quickly recreate environments.

Example:

A developer wants to deploy the same web application to multiple environments (Development, Testing, and Production). Instead of manually creating resources each time, they create a Bicep template and deploy it with a single Azure CLI command, ensuring identical infrastructure across all environments.

Lab Steps

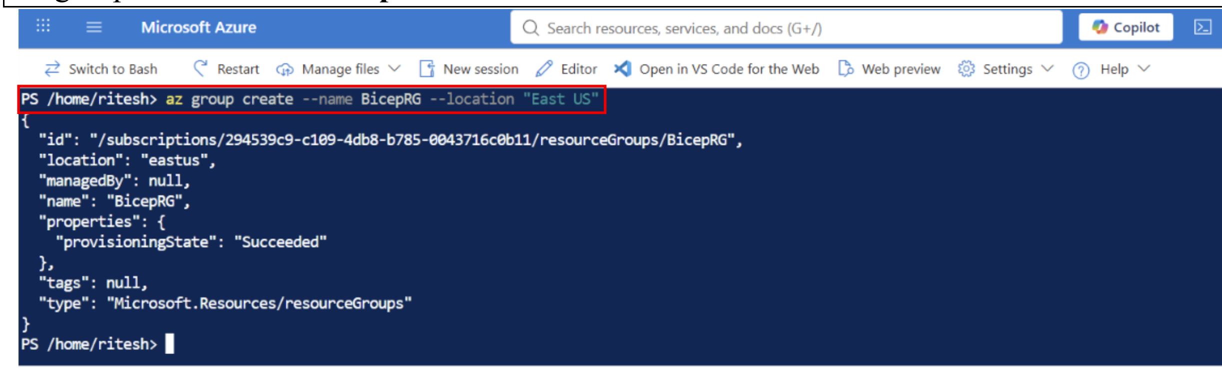
Step 1: Open Azure Cloud Shell

- Sign in to the Azure Portal.
- Open **Azure Cloud Shell**.
- Select **PowerShell** or **Bash**.

Step 2: Create a Resource Group

- Create a new Resource Group **BicepRG**.

```
az group create --name BicepRG --location "East US"
```

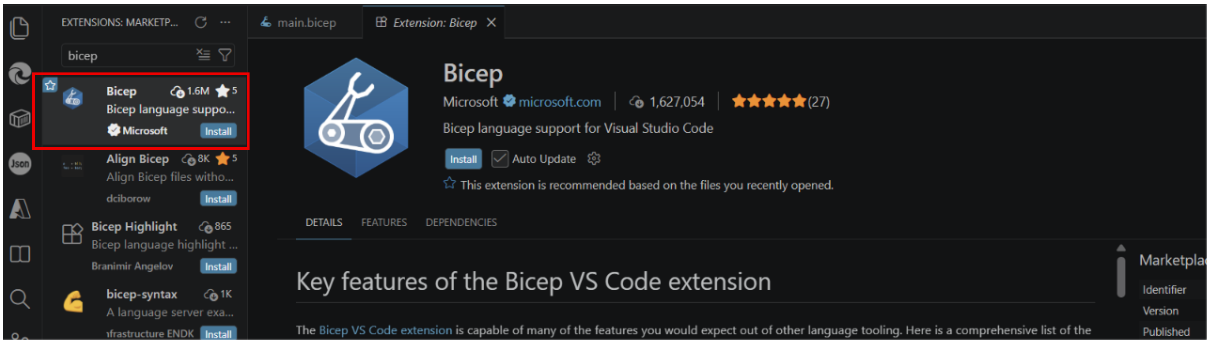


Step 3: Create a Project Folder

- Create a working directory **Bicep webapp**.
- Navigate to the directory.

Step 4: Create a Bicep File

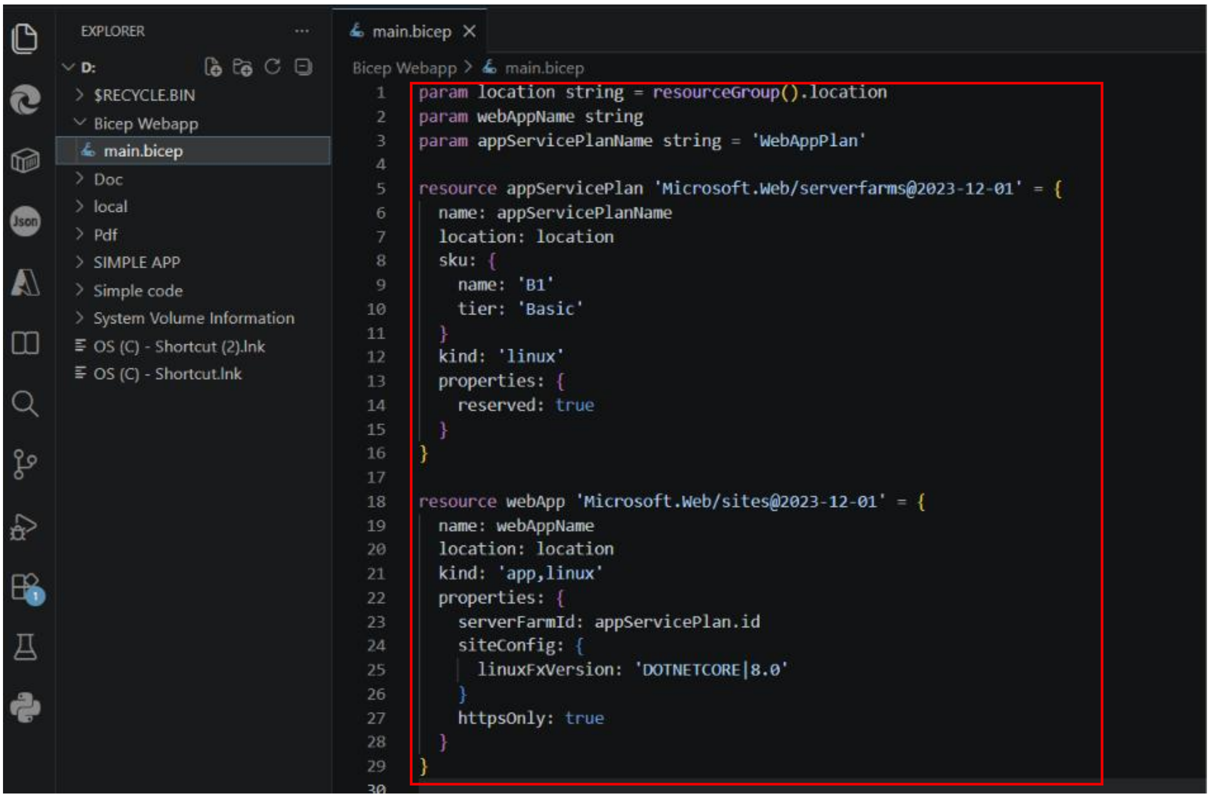
- Go the **VS Code**
- Open the **Bicep** directory
- Create a file named **main.bicep**.
- Install **Bicep extension**



Step 4: Add the Bicep Template

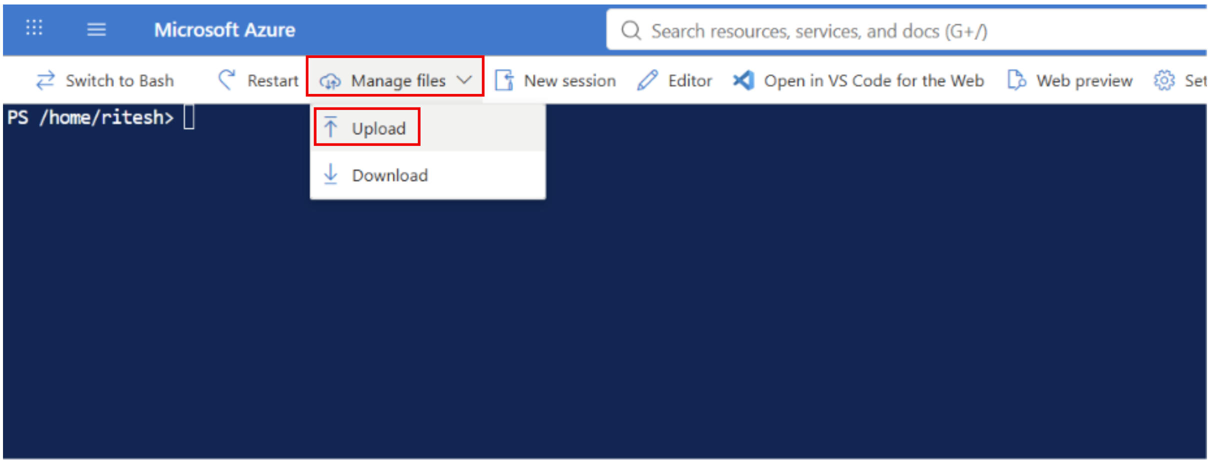
- Open **main.bicep** and add the following code.

```
param location string = resourceGroup().location
param webAppName string
param appServicePlanName string = 'WebAppPlan'

resource appServicePlan 'Microsoft.Web/serverfarms@2023-12-01' = {
  name: appServicePlanName
  location: location
  sku: {
    name: 'B1'
    tier: 'Basic'
  }
  kind: 'linux'
  properties: {
    reserved: true
  }
}

resource webApp 'Microsoft.Web/sites@2023-12-01' = {
  name: webAppName
  location: location
  kind: 'app,linux'
  properties: {
    serverFarmId: appServicePlan.id
    siteConfig: {
      linuxFxVersion: 'DOTNETCORE|8.0'
    }
    httpsOnly: true
  }
}
```

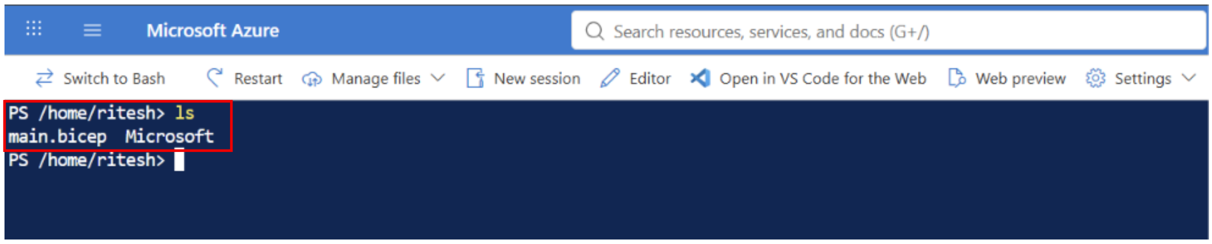


Step 5: Upload Bicep file

- Go to **Azure power shell**.
- Navigate to **Manage file.**
- Click on the **upload**.
- Upload the **main.bicep** file



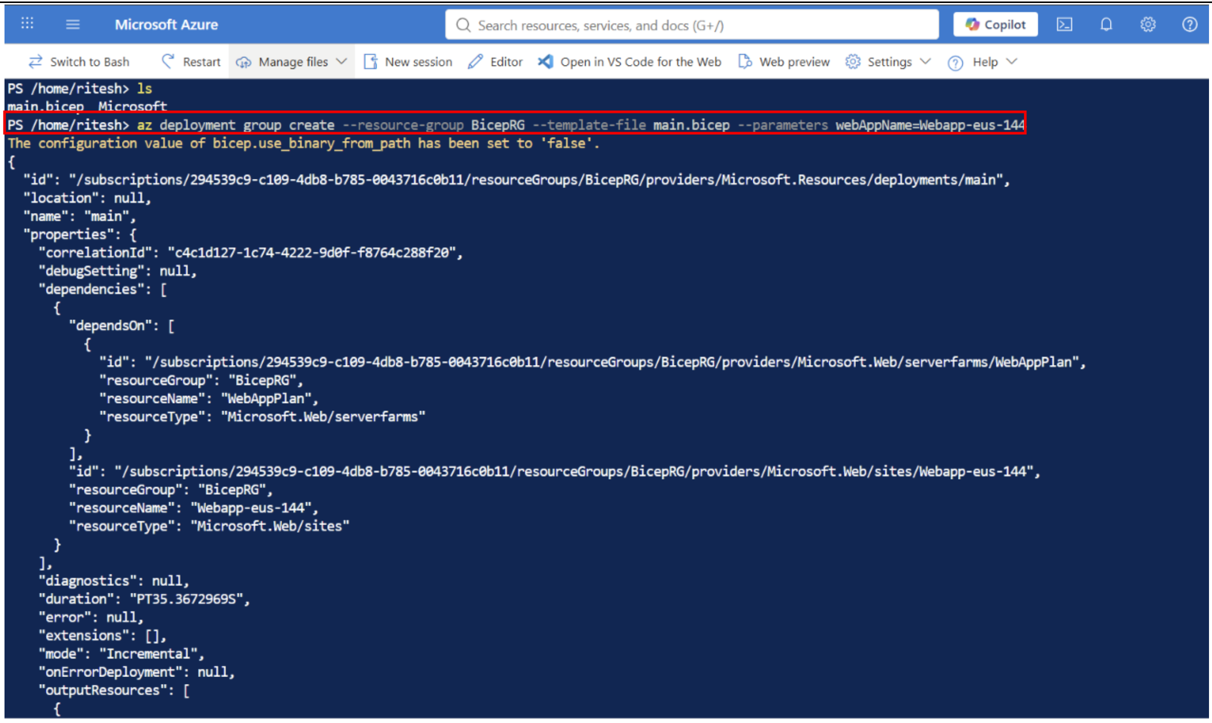
- Run **ls** command.
- **Main.bicep** file is uploaded



Step 6: Deploy the Bicep Template

- Deploy the template using Azure CLI.
- Webapp name is **Webapp-eus-144**.
- Run this command.

az deployment group create --resource-group BicepRG --template-file main.bicep --parameters webAppName=**Webapp-eus-144**



Step 7: Verify the Deployment

- Open the Azure Portal.
- Navigate to the **BicepRG** Resource Group.
- Verify that the following resources were created:
  - App Service Plan
  - Azure Web App